applications that leverage efficient nearest neighbor search as part of their workflow, like *clustering*.

8.1 The Nearest Neighbors Search Problem

Let’s start our journey for this chapter from figure 8.1: a map showing a few cities (taken from an alternate universe) and the locations of some warehouses in their area.

Imagine you are living in 1990s, the dawn of the Internet era, with e-commerce moving its very first steps.

You are running an online store where you sell locally produced products by collaborating with a few retailers – they sell to shops in the real world, and you provide them with the infrastructure to also sell online, for a small price.

Each warehouse will take care of deliveries for their orders, but to gain traction and have more retailers hop on board, you offer a special deal: for each delivery further than 10 km away, you will cut your commission proportionally to the distance.